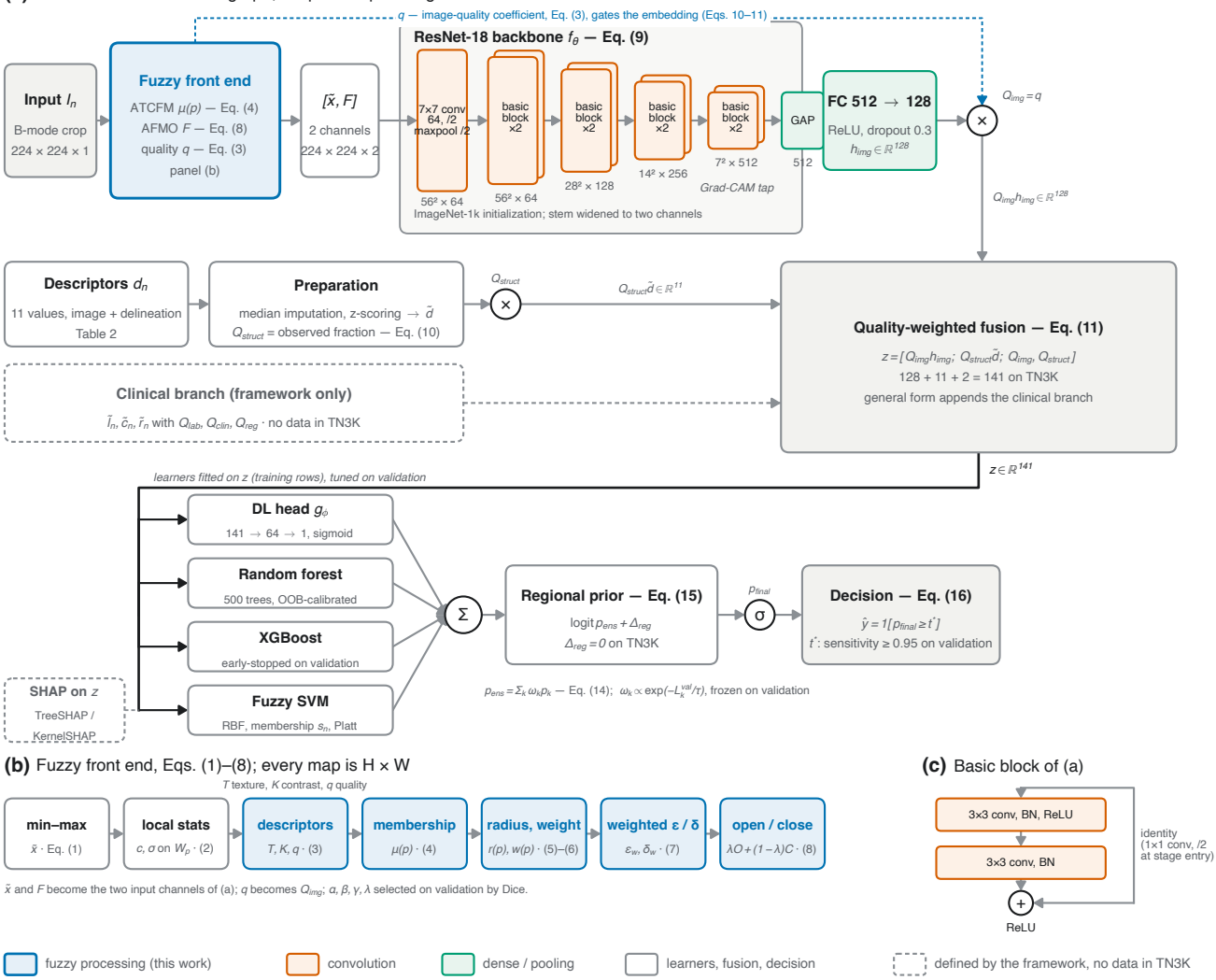

(a) TAMES AI as a data-flow graph; shapes are per image
q — image-quality coefficient, Eq. (3), gates the embedding (Eqs. 10–11)
Input I_n
B-mode crop
224 × 224 × 1
Fuzzy front end
ATCFM $\mu(p)$ — Eq. (4)
AFMO F — Eq. (8)
quality q — Eq. (3)
panel (b)
$[\tilde{x}, F]$
2 channels
224 × 224 × 2
ResNet-18 backbone f_θ — Eq. (9)
7×7 conv 64, /2
maxpool /2
basic block ×2
basic block ×2
basic block ×2
basic block ×2
GAP
512
56² × 64
56² × 64
28² × 128
14² × 256
7² × 512
Grad-CAM tap
ImageNet-1k initialization; stem widened to two channels
FC 512 → 128
ReLU, dropout 0.3
$h_{img} \in \mathbb{R}^{128}$
$Q_{img} = q$
$Q_{img} h_{img} \in \mathbb{R}^{128}$
Descriptors d_n
11 values, image + delineation
Table 2
Preparation
median imputation, z-scoring → $\tilde{d}$
Q_{struct} = observed fraction — Eq. (10)
Q_{struct}
$Q_{struct} \tilde{d} \in \mathbb{R}^{11}$
Clinical branch (framework only)
$\tilde{l}_n, \tilde{c}_n, \tilde{r}_n$ with $Q_{lab}, Q_{clin}, Q_{reg}$ · no data in TN3K
Quality-weighted fusion — Eq. (11)
$z = [Q_{img} h_{img};\ Q_{struct} \tilde{d};\ Q_{img}, Q_{struct}]$
128 + 11 + 2 = 141 on TN3K
general form appends the clinical branch
$z \in \mathbb{R}^{141}$
learners fitted on z (training rows), tuned on validation
DL head g_φ
141 → 64 → 1, sigmoid
Random forest
500 trees, OOB-calibrated
XGBoost
early-stopped on validation
Fuzzy SVM
RBF, membership s_n, Platt
SHAP on z
TreeSHAP / KernelSHAP
Σ
Regional prior — Eq. (15)
logit $p_{ens} + \Delta_{reg}$
$\Delta_{reg} = 0$ on TN3K
p_{final}
σ
Decision — Eq. (16)
$\hat{y} = 1[p_{final} \geq t^*]$
t^*: sensitivity ≥ 0.95 on validation
$p_{ens} = \Sigma_k \omega_k p_k$ — Eq. (14); $\omega_k \propto \exp(-L_k^{val}/\tau)$, frozen on validation
(b) Fuzzy front end, Eqs. (1)–(8); every map is H × W
T texture, K contrast, q quality
min–max
$\tilde{x}$ · Eq. (1)
local stats
c, σ on W_p · (2)
descriptors
T, K, q · (3)
membership
$\mu(p)$ · (4)
radius, weight
$r(p), w(p)$ · (5)–(6)
weighted ε / δ
ε_w, δ_w · (7)
open / close
$\lambda O + (1-\lambda) C$ · (8)
$\tilde{x}$ and F become the two input channels of (a); q becomes Q_{img}; $\alpha, \beta, \gamma, \lambda$ selected on validation by Dice.
(c) Basic block of (a)
3×3 conv, BN, ReLU
3×3 conv, BN
identity (1×1 conv, /2 at stage entry)
ReLU
fuzzy processing (this work)
convolution
dense / pooling
learners, fusion, decision
defined by the framework, no data in TN3K